Answer key

Practice Test 3 (State/National level) · 50 questions · 36 with checked app keys

Each answer lists short steps, the vault note (Question Types folder) and the app keys. Every key line was run in the app and its screen value matched the computed answer (check_keys.py). Accept numeric answers within about 2% with the right unit.

Multiple-choice answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
C	D	C	B	D	D	C	D	D	A	C	C	C	A

Section I: Multiple choice

1. C. 1845

1. Kirchhoff stated the laws in 1845, aged about 21.
2. They extend Ohm's law (1827) to networks with many loops.
3. 1785 is Coulomb's law; 1887 is Hertz's work.

Note: A1 History/Kirchhoff
App: no app: fact or concept

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2. D. It used abstract mathematics to describe a physical experiment

1. Many German physicists of the time preferred experiment to mathematical theory.
2. Ohm described the circuit with mathematics, so they rejected it at first.

Note: A1 History/Ohm
App: no app: fact or concept

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3. C. Made the spark jump more easily

1. UV light on the gap let the spark jump more easily.
2. This was the first observation of the photoelectric effect.

Note: A1 History/Hertz
App: no app: fact or concept

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4. B. A torsion balance

1. A torsion balance: a thin wire twists when a force acts on a needle.
2. The twist angle measures very small forces.

Note: A1 History/Coulomb
App: no app: fact or concept

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5. D. On dry winter days

1. Humid air lets charge leak away.
2. Dry air (winter heating) lets charge build up.

Note: A2 Static Electricity and Coulomb/Static Hazards
App: no app: fact or concept

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6. D. A large current flows to ground and trips the breaker (or blows the fuse)

1. The ground wire is a low-resistance path from the case.
2. Hot to case to ground is almost a short circuit.
3. The large fault current trips the breaker, so the case does not stay live.

Note: A3 DC and AC/Household Wiring
App: no app: fact or concept

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7. C. 4

1. Leading zeros do not count.
2. 4, 0, 5 and the trailing 0 after the decimal point count.
3. So 4 significant figures.

Note: A4 Current Voltage Resistance Power Energy/Units Prefixes and Sig Figs
App: no app: fact or concept

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8. D. It trips, because 9 mA leaks to ground

1. Leak = 7.500 − 7.491 = 0.009 A = 9 mA.
2. A Class A GFCI must trip at 6 mA or more.
3. 9 mA ≥ 6 mA, so it trips.

Note: A6 Switches Relays Fuses Breakers GFCI/GFCI
App: no app: fact or concept

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9. D. XOR

1. (0,0) off; one flip gives on; a second flip gives off again.
2. On when the switches differ: XOR.

Note: A6 Switches Relays Fuses Breakers GFCI/3-Way and 4-Way Switching
App: `t0110` → `This is an X0R gate` *(checked)*

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10. A. A slow-blow (time-delay) fuse of the same rating

1. A cold filament has a much lower resistance, so the switch-on (inrush) current is large.
2. A slow-blow fuse rides through a short inrush.
3. It still protects at the same rating.

Note: A6 Switches Relays Fuses Breakers GFCI/Fuses
App: no app: fact or concept

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11. C. 3

1. NOR(A,A) = A'. NOR(B,B) = B'.
2. NOR(A', B') = (A' + B')' = AB.
3. 3 gates.

Note: A8 Digital Logic/Universal Gates (NAND and NOR)
App: no app: fact or concept

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12. C. A + B + C′

1. De Morgan: (XY)' = X' + Y'.
2. F = ((A+B)')' + C' = A + B + C'.

Note: A8 Digital Logic/De Morgan and Bubble Pushing
App: `t-(-(a+b)c)` → `F = A + B + C'` *(checked)*

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13. C. 45 Ω

1. Face diagonal of a cube = 3R/4 (edge = 7R/12, body diagonal = 5R/6).
2. 3 × 60 / 4 = 45 ohm.
3. Symmetry: corners at the same voltage can be joined.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N1–2–3–4–1 60 5–6–7–8–5 1–5 2–6 3–7 4–8 =1–3` → `R(1–3) = 45 ohm` *(checked)*

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14. A. Only the red LED lights; it gets about 20 mA

1. The red LED holds the shared node at about 1.8 V.
2. 1.8 V is below 3.1 V, so the blue LED stays dark.
3. I = (9.0 – 1.8)/360 = 20 mA, all in the red LED.
4. That is why each LED needs its own resistor.

Note: A7 Diodes and LEDs/LEDs
App: `l s9 f1.8 r360` → `i LED current = 20 mA` *(checked)*

Section II: Short answer and calculations

15. 550 mN, toward +x (toward the +3.0 µC charge)

1. Pull toward the +4.0 µC: k(4.0e-6)(2.0e-6)/0.30^2 = 800 mN (–x).
2. Pull toward the +3.0 µC: k(2.0e-6)(3.0e-6)/0.20^2 = 1.3 N (+x).
3. Net = 1.348 N – 799.1 mN = 550 mN toward +x.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `Q 4u 0, −2u .3, 3u .5` → `F on q2 (−2 uC at 0.3): 549.2 mN right` *(checked)*

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16. x = 0.60 m

1. Between opposite charges both fields point the same way, so E = 0 is outside.
2. It is on the side of the SMALLER charge: x > 0.30 m.
3. $4/x^2 = 1/(x – 0.30)^2$, so 2(x – 0.30) = x.
4. x = 0.60 m.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: `Q 4u 0, −1u .3` → `E = 0 at x = 0.6 m` *(checked)*

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17. 1.0 N (F_x = +600 mN, F_y = –840 mN)

1. q_2 attracts q_1 toward +x: k(3.0e-6)(2.0e-6)/0.30^2 = 600 mN.
2. q_3 repels q_1 toward –y: k(3.0e-6)(5.0e-6)/0.40^2 = 840 mN.
3. The two parts are perpendicular: use Pythagoras.
4. |F| = $\sqrt{(0.5993^2 + 0.8428^2)}$ = 1.0 N.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `Q 3u 0 0, −2u .3 0, 5u 0 .4` → `F on q1: 1.034 N at −54.58 deg` *(checked)*

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18. 3.6 N, repel (each +3.0 μC)

1. Identical spheres share the total charge equally.
2. Each: (8.0 – 2.0)/2 = +3.0 μC.
3. F = k(3.0e-6)^2/0.15^2 = 3.6 N, repel.

Note: A2 Static Electricity and Coulomb/Charge Basics
App: `q a8u b-2u r.15` → `If they touch: each 3 uC, |F| x 0.5625, now REPEL` *(checked)*
App: `q a3u b3u r.15` → `f Force (+ repel) = 3.595 N` *(checked)*

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19. 1 Ω

1. I = 8.2 V / 10.0 ohm = 820 mA.
2. Lost volts = 9.0 – 8.2 = 0.8 V (1 significant figure).
3. r = 0.8 / 0.82 = 1 Ω (about 0.98 ohm).

Note: A5 DC Circuits/Internal Resistance and Power Transfer
App: `i e9 v8.2 l10` → `r Internal r = 0.97561 ohm` *(checked)*

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20. 21 W

1. Maximum power transfer: load R = internal r.
2. $P_{max} = E^2/(4r)$.
3. 81 / (4 × 0.9756) = 21 W.

Note: A5 DC Circuits/Internal Resistance and Power Transfer
App: `i e9 v8.2 l10` → `x Max power (load = r) = 20.76 W` *(checked)*

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21. 15 V; 77 Wh

1. Series adds voltage: 4 × 3.7 = 14.8 V.
2. Parallel adds capacity: 2 × 2600 = 5200 mAh.
3. Energy = 14.8 V × 5.2 Ah = 77 Wh.

Note: A3 DC and AC/Batteries and Cells
App: `p s4 p2 v3.7 c2600` → `w Pack energy = 76.96 Wh` *(checked)*
App: `p s4 p2 v3.7 c2600` → `a Pack voltage = 14.8 V` *(checked)*

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22. 651 V

1. $V_{peak} = \sqrt{2} \times 230 = 325$ V.
2. $V_{pp} = 2 \times V_{peak} = 651$ V.

Note: A3 DC and AC/AC Characteristics
App: `w r230` → `a Peak-to-peak = 650.5 V` *(checked)*

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23. 64 °C

1. $R = R_0(1 + \alpha\Delta T)$.
2. ΔT = (6.10/5.20 – 1)/0.0039 = 44.4 °C.
3. T = 20 + 44.4 = 64 °C.

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: `R o5.2 r6.1 a.0039` → `t Temperature T = 64.379 C` *(checked)*

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24. $139.61 (930.75 kWh saved; about $140)

1. Power saved = 10 × (60 – 9.0) = 510 W.
2. Energy saved = 0.510 kW × 5.0 h × 365 = 930.75 kWh.
3. Money = 930.75 × $0.15 = $139.61.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `e p510 h5 d365 r.15` → `c Cost = $139.61 (139.61)` *(checked)*

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25. 1.26 times

1. R = ρL/A, so same R and L needs A proportional to ρ.
2. A_{Al}/A_{Cu} = 2.65/1.68 = 1.577.
3. A is proportional to d^2, so d ratio = √1.577 = 1.256.

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: by hand (ratio; the r tool does one wire at a time)

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26. 4.80 V

1. The meter is in parallel with the bottom resistor: 100k || 200k = 66.67 kΩ.
2. Divider: 12.0 × 66.67k / (100k + 66.67k) = 4.80 V.
3. The meter loads the circuit and reads low.

Note: A5 DC Circuits/Meters and Measurement
App: `d v12 a100k b100k/200k` → `o Vout (across R2) = 4.8 V` *(checked)*

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27. 46.1 Ω

1. The ladder looks the same after one step: R = 20 + (60 || R).
2. R – 20 = 60R/(60 + R), so R^2 – 20R – 1200 = 0.
3. R = (20 + √(400 + 4800))/2 = 46.1 Ω (take the positive root).

Note: A5 DC Circuits/Hard Resistor Networks
App: `L a20 b60` → `r Req (a first) = 46.06 ohm` *(checked)*

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28. 240 Ω

1. Total R = 9.0 / 0.025 = 360 ohm.
2. Parallel pair = 360 – 220 = 140 ohm.
3. 1/R = 1/140 – 1/330, so R = 240 Ω (2 significant figures, from 9.0 V and 25 mA).

Note: A5 DC Circuits/Series and Parallel Resistance
App: `N V1–0 9 1–2 220 2–0 330 2–0 x I=25m` → `x = 243.2 ohm (R2–0)` *(checked)*

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29. The 40 W bulb; 9.37 W

1. R = V^2/P: 360, 240 and 144 ohm.
2. Series: I = 120 / 744 = 161 mA.
3. Same current: the biggest R (the 40 W bulb) uses the most power.
4. P = I^2R = 0.1613^2 × 360 = 9.37 W.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `B40 120, 60 120, 100 120 =120s` → `bulb 1: 58.06 V, P = 9.365 W` *(checked)*

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30. 2.25 times (it gets brighter)

1. Before: bulb 2 gets V/3 (the pair is R/2 of a total 3R/2).
2. After: bulbs 1 and 2 are in series, so bulb 2 gets V/2.
3. P is proportional to V^2: $(1/2)^2 / (1/3)^2 = 9/4 = 2.25$.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `NV1-0 1 1-2 1 2-0 1 o2-0 1` → `2-0 1: 111.1 mW -> 250 mW x2.25 brighter` *(checked)*

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31. 20 A (12.5 A load)

1. I = 1500 / 120 = 12.5 A.
2. Continuous load: I must be at most 80% of the rating.
3. 15 A × 0.8 = 12 A, too small. 20 A × 0.8 = 16 A, OK. So 20 A.

Note: A6 Switches Relays Fuses Breakers GFCI/Circuit Breakers
App: `u p1500 v120` → `Continuous (x1.25 = 15.62 A): 20 A` *(checked)*
App: `u p1500 v120` → `i Current = 12.5 A` *(checked)*

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32. 6.3 mA

1. The diode is forward biased (12 V > 5 V).
2. KVL: 12.0 – 0.7 – I(1000) – 5.0 = 0.
3. I = 6.3 V / 1000 ohm = 6.3 mA.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `l s12-5 f.7 r1k` → `i LED current = 6.3 mA` *(checked)*

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33. 530 mV

1. ΔT = 85 – 25 = 60 °C.
2. Drop = 60 × 0.0020 = 0.12 V.
3. $V_F = 0.650 - 0.12 = 0.53$ V.

Note: A7 Diodes and LEDs/PN Junction and I-V Curve
App: by hand

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34. F = B′C′ + D′

1. All even minterms (D = 0) are 1 or don't care: group of 8 gives D'.
2. 0, 1, 8, 9 (B = 0, C = 0) give the quad B'C' (1 is a don't care).
3. F = B'C' + D'.

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `t1x10x01011xx1010` → `F = B'C' + D'` *(checked)*

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35. S = 0, C_{out} = 1

1. 1 + 0 + 1 = 2 = binary 10.
2. S is the low bit: 0.
3. C_{out} is the high bit: 1.

Note: A8 Digital Logic/Adders
App: by hand (or x tool: 1+0+1 in binary is 10)

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36. F = AB + AC + BC

1. Truth table: F = 1 for 011, 101, 110, 111.
2. Column 00010111.
3. K-map: three pairs give AB + AC + BC.

Note: A8 Digital Logic/Logic Design Word Problems
App: `t00010111` → `F = AB + AC + BC` *(checked)*

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37. 100101000 = 296

1. 10111011 = 187, 1101101 = 109.
2. 187 + 109 = 296.
3. 296 = 100101000 (256 + 32 + 8).

Note: A8 Digital Logic/Binary Numbers and Addition
App: `x10111011+1101101` → `sum bin 100101000` *(checked)*

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38. 2.53 kΩ

1. Voltage across the known R = 9.00 − 3.15 = 5.85 V.
2. Same current: $R_x/R_k = V_x/V_k$.
3. R_x = 4.70k × 3.15/5.85 = 2.53 kΩ.

Note: A9 Hands-On and Meters/Task - Mystery Resistor
App: `m a4.7k v5.85 w3.15` → `x Mystery R (Rx) = 2.531 kohm` *(checked)*

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39. 1.3 h (about 78 min)

1. Battery energy = 5.0 Ah × 3.85 V = 19.25 Wh.
2. Useful power = 9.0 × 2.0 × 0.82 = 14.76 W.
3. t = 19.25 / 14.76 = 1.3 h.

Note: A9 Hands-On and Meters/Task - USB Charger
App: `z c5000 v3.85 s9 a2 e.82` → `t Charge time = 1.3042 h` *(checked)*

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40. 470 || (220 + 330) = 253.4 Ω

1. 220 + 330 = 550 ohm.
2. 470 || 550 = 470 × 550 / 1020 = 253.4 ohm.
3. Error = +0.17%: inside 1%.

Note: A9 Hands-On and Meters/Task - Target Resistance
App: `h 100 220 330 470=253` → `253.4 (+0.171%) 470||(330+220)` *(checked)*
App: `s470/(220+330)` → `Req = 253.4 ohm` *(checked)*

Section III: Circuit problems

Problem A: two sources (Kirchhoff) (Questions 41–44)

In the circuit below, E1 = 15.0 V and E2 = 6.0 V are ideal batteries (the + side is the long plate, at the top of each). R1 = 5.0 Ω, R2 = 3.0 Ω, R3 = 10 Ω. Let I_1 be the current in R1 flowing toward node A and I_2 the current in R2 flowing toward node A.

41. 1.42 A

1. KCL: the current in R3 (A to B) is $I_1 + I_2$.
2. Left loop: $15 = 5\ I_1 + 10(I_1 + I_2)$, so $15\ I_1 + 10\ I_2 = 15$.
3. Right loop: $6 = 3\ I_2 + 10(I_1 + I_2)$, so $10\ I_1 + 13\ I_2 = 6$.
4. Solve: $I_1 = 135/95 = 1.42$ A.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a15 b10 c15 d10 e13 f6` → `x x (I1 or V1) = 1.4211` *(checked)*

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42. I_2 = -632 mA: 632 mA flows from A into the + side of E2 (E2 is being charged)

1. From the same equations: $I_2 = -60/95 = -632$ mA.
2. The minus sign means the real current flows away from A, into the + terminal of E2.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a15 b10 c15 d10 e13 f6` → `y y (I2 or V2) = −0.63158` *(checked)*
App: `N V1–0 15 1–2 5 V3–0 6 3–2 3 2–0 10` → `V3–0 6: 631.6 mA into +, takes 3.789 W` *(checked)*

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43. 7.89 V

1. $V_A = R3(I_1 + I_2) = 10 \times 0.7895 = 7.89$ V.
2. Check with node analysis: $(V-15)/5 + (V-6)/3 + V/10 = 0$ gives $V = 150/19$.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 15 1–2 5 V3–0 6 3–2 3 2–0 10` → `V1 = 15 V V2 = 7.895 V` *(checked)*

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44. 3.79 W

1. $P = E2 \times |I_2| = 6.0 \times 0.6316 = 3.79$ W.
2. Check: E1 gives $15 \times 1.421 = 21.3$ W = resistors 17.5 W + E2 3.79 W.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 15 1–2 5 V3–0 6 3–2 3 2–0 10` → `V3–0 6: 631.6 mA into +, takes 3.789 W` *(checked)*

Problem B: unbalanced bridge (Questions 45–46)

A 20.0 V ideal battery feeds the bridge below. Node P is between 150 Ω and 450 Ω; node Q is between 300 Ω and 200 Ω. A 100 Ω resistor R_{PQ} joins P and Q.

45. 7.00 V

1. Each side is a divider.
2. $V_P = 20 \times 450/600 = 15.00$ V.
3. $V_Q = 20 \times 200/500 = 8.00$ V.
4. Difference = 7.00 V.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N V1–0 20 1–2 150 1–3 300 2–0 450 3–0 200 V2–3` → `V2 – V3 = 7 V` *(checked)*

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46. 21.1 mA from P to Q

1. Node P: $(V_P - 20)/150 + V_P/450 + (V_P - V_Q)/100 = 0$.
2. Node Q: $(V_Q - 20)/300 + V_Q/200 + (V_Q - V_P)/100 = 0$.
3. Solve the two equations: $V_P = 12.632$ V, $V_Q = 10.526$ V.
4. I = (12.632 − 10.526)/100 = 21.1 mA, from P to Q.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 20 1–2 150 1–3 300 2–0 450 3–0 200 2–3 100` → `R2–3 100: 2.105 V 21.05 mA 2>3 44.32 mW` *(checked)*
App: `j a17 b–9 c120 d–6 e11 f40` → `x x (I1 or V1) = 12.632` *(checked)*

Problem C: logic circuit (Questions 47–48)

The circuit below has three NAND gates, inputs A, B, C, D and output F.

47. F = AB + CD

1. F = ((AB)'(CD)')'.
2. De Morgan: F = (AB)'' + (CD)'' = AB + CD.
3. NAND-NAND = AND-OR.

Note: A8 Digital Logic/De Morgan and Bubble Pushing
App: `tab+cd` → `F = AB + CD` *(checked)*

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48. 7 rows

1. AB = 1: 4 rows. CD = 1: 4 rows.
2. Both: 1 row (1111) counted twice.
3. 4 + 4 − 1 = 7.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `tab+cd` → `m(3,7,11,12,13,14,15)` *(checked)*

Problem D: diode logic (Questions 49–50)

Inputs A and B are each either 0 V or 5.0 V. D1 and D2 are silicon diodes (0.7 V when on). R = 10 kΩ goes from the output to ground.

49. OR gate

1. Any input at 5 V turns its diode on and pulls the output high.
2. Both inputs at 0 V: no current, output = 0 V.
3. So the output is high if A OR B is high.

Note: A7 Diodes and LEDs/Diode Applications
App: no app: fact or concept

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50. 4.3 V; 430 µA

1. D1 is on: the output = 5.0 − 0.7 = 4.3 V.
2. D2 is reverse biased (4.3 V on its cathode, 0 V on its anode).
3. I = 4.3 V / 10 k = 430 µA.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `l s5 f.7 r10k` → `i LED current = 430 uA` *(checked)*
